

Scales can add observations, cannot create missing directions

R0.70M left a natural hope: a single-scale covariance may be rank-deficient, but a sum of outer products across adjacent scales, multiple centers, or a time window might automatically form a three-dimensional frame. R0.70N gives a negative answer. With nonnegative weights and scalar or componentwise filters, if all observations share one null vorticity direction, their sum retains it; smooth unforced periodic Navier–Stokes solutions already provide exact witnesses.

Status • R0.70N exact certificate, independent internal review PASS, and formal figure completed

Universal multiscale coercivity route closed

Version v0.70N • 2026-08-25

Exact generators: all 4 groups passed

Figure data: 553 rows of closed-form data

Formal figure package: PDF / SVG / 600 dpi PNG

Next gate: spectral stratification into coercive, near-plane, and near-line regimes

00 / EXACT CONCLUSION

No universal positive frame constant exists

For a family of filtered vorticities Ω_j , nonnegative weights w_j , and nonnegative spatial weights χ_j , define

$$Q_j = \int \chi_j \Omega_j \otimes \Omega_j dx, \quad Q_k = \sum_j w_j Q_j.$$

The original candidate requires a solution-independent $c > 0$ such that every smooth solution satisfies

$$Q_k \succeq c \operatorname{tr}(Q_k) I.$$

This statement is false. An exact periodic shear leaves every Q_j with the same single vorticity direction; a single-axis helical wave supplies a common normal direction; and genuinely three-dimensional, full-rank whole-space Schwartz data can still make the optimal constant tend to zero.

The no-go theorem established at this stage. Nonnegative aggregation over scales, centers, or time cannot provide a uniform positive frame constant for scalar-filtered vorticity across all smooth periodic Navier–Stokes solutions. The conclusion covers only the observation class stated here; it does not exclude additional directional-excitation hypotheses or vector-valued mixing observations.

01 / AGGREGATE LEDGER

Summation retains every source mismatch and creates no damping

If the per-scale ledgers are written as

$$\dot{Q}_j = \Sigma_j Q_j + Q_j \Sigma_j + F_j,$$

After choosing a reference strain Σ_* , the aggregate satisfies the exact identity

$$\dot{Q}_k = \Sigma_* Q_k + Q_k \Sigma_* + \mathcal{F}_k,$$

$$\mathcal{F}_k = \sum_j w_j [F_j + (\Sigma_j - \Sigma_*) Q_j + Q_j (\Sigma_j - \Sigma_*)] + \sum_j \dot{w}_j Q_j.$$

Consequently, scale-source mismatch, moving weights, cutoff, viscosity, and commutator terms must all be retained. Pulling back along $\dot{G}_* = \Sigma_* G_*$ yields only

$$\dot{\hat{Q}}_k = G_*^{-1} \mathcal{F}_k G_*^{-\top},$$

This does not automatically make these terms dissipative.

02 / OPTIMAL FRAME CONSTANT

The optimal constant is the normalized smallest eigenvalue

For any nonzero positive-semidefinite aggregate, the optimal constant is

$$c_*(Q_k) = \frac{\lambda_{\min}(Q_k)}{\text{tr } Q_k} = \inf_{|v|=1} \frac{\sum_j w_j \int \chi_j |v \cdot \Omega_j|^2}{\sum_j w_j \int \chi_j |\Omega_j|^2}.$$

Nonnegativity also yields the exact kernel formula

$$\ker \mathcal{Q}_k = \bigcap_{j:w_j>0} \ker \mathcal{Q}_j.$$

This formula already identifies the key mechanism: summation can reinforce existing directions, but it cannot remove a null direction shared by every positively weighted observation.

03 / EXACT PERIODIC SHEAR

The simplest smooth unforced solution already preserves a common rank-one range

On the three-dimensional torus, take

$$u_s = Ae^{-\nu N^2 t} \sin(Ny)e_1, \quad \omega_s = -ANe^{-\nu N^2 t} \cos(Ny)e_3.$$

It satisfies $(u_s \cdot \nabla)u_s = 0$ and $\partial_t u_s = \nu \Delta u_s$. Any scalar or componentwise filter that retains this Fourier mode changes only its scalar amplitude, not the e_3 direction; every nonnegative sum over scales, centers, and times therefore remains

$$\mathcal{Q}_s = q_s e_3 e_3^\top, \quad \lambda_{\min}(\mathcal{Q}_s) = 0, \quad \text{tr } \mathcal{Q}_s = q_s > 0.$$

Therefore, increasing the number of observations does not change this counterexample.

04 / HELICAL BOUNDARY AND CONTROL

The direction may rotate and still span only a plane

Exact single-axis helical wave

$$u_b = Ae^{-\nu N^2 t} (\cos Nz, -\sin Nz, 0), \quad \nabla \times u_b = Nu_b,$$

Its direction rotates with z but remains in $e_3^\perp$, so the aggregated covariance still has the common kernel $\text{span}(e_3)$. This also shows that directional variation alone does not constitute a three-dimensional frame.

A positive control must genuinely add a new axis. Let

$$b_z = (\sin z, \cos z, 0), \quad b_x = (0, \sin x, \cos x), \quad u_{2h} = e^{-\nu t} (ab_z + bb_x).$$

For full-torus, translation-invariant Fourier multipliers, one obtains

$$\mathcal{Q}_{2h} = \text{diag}(\alpha, \alpha + \beta, \beta), \quad \det \mathcal{Q}_{2h} = \alpha\beta(\alpha + \beta),$$

It is positive definite only when $\alpha, \beta > 0$, in which case $c_* = \min(\alpha, \beta)/[2(\alpha + \beta)]$. This is a conditional positive example and cannot be generalized into a statement about all Beltrami fields.

05 / GENUINE 3D AND FULL SPACE

Even after excluding exactly two-dimensional cases or requiring full rank, there is still no uniform constant

For any fixed finite observation scheme, add to the shear a genuinely three-dimensional divergence-free perturbation of size ε . The transverse Rayleigh quotient is $O(\varepsilon^2)$, while the trace retains a positive limit; hence

$$\frac{\lambda_{\min}(\mathcal{Q}^\varepsilon)}{\text{tr } \mathcal{Q}^\varepsilon} \rightarrow 0.$$

This step does not claim that an arbitrary filter frame is full rank. For an explicit positive-definite calibration, use only the full-torus identity filter and take the three-dimensional perturbation to be $v_0 = b_z + b_x$; then the covariance is positive definite for every $\varepsilon > 0$, while the optimal constant still tends to zero.

The whole space also admits a closed-form Schwartz family. Take

$$\psi_L = e^{-(x^2+y^2)/2-z^2/(2L^2)}, \quad u_L = (-y\psi_L, x\psi_L, 0),$$

Then the initial covariance is strictly positive definite, and

$$Q_L = \pi^{3/2} \text{diag} \left(\frac{1}{4L}, \frac{1}{4L}, 2L \right), \quad c_*(Q_L) = \frac{1}{8L^2 + 2} \rightarrow 0.$$

Therefore, genuine three-dimensionality, finite energy, and full rank are all insufficient to provide a data-independent quantitative lower bound.

06 / COMMON-SUBSPACE THEOREM

For nonlocal filters, the subspace condition must be imposed on the entire input domain

If the original vorticity takes values in a fixed proper subspace V throughout the entire spatial domain actually read by each nonlocal filter and at every observed time, then scalar or componentwise filtering commutes with constant projections, so every output still lies in V . A weaker direct alternative is to require each filtered output itself to lie in V .

$$\text{Ran } \mathcal{Q} \subset V, \quad V^\perp \subset \ker \mathcal{Q}.$$

The argument that no term can cancel a square is used only for nonnegative spatial and temporal weights. Signed weights, matrix-valued filters that rotate components, or additional observations change the problem and require realizability and the energy ledger to be established again.

07 / JOURNAL FIGURE

Rank boundaries, conditional positive examples, and the absence of a uniform constant

Scalar multi-scale vorticity frames: exact nullspaces and the minimal escape

common-subspace preservation · two-axis directional balance · full-rank constants can still degenerate



EXACT PDE WITNESSES + ANALYTIC FRAME IDENTITIES / ROUTE NO-GO, NOT A REGULARITY OR MILLENNIUM RESULT

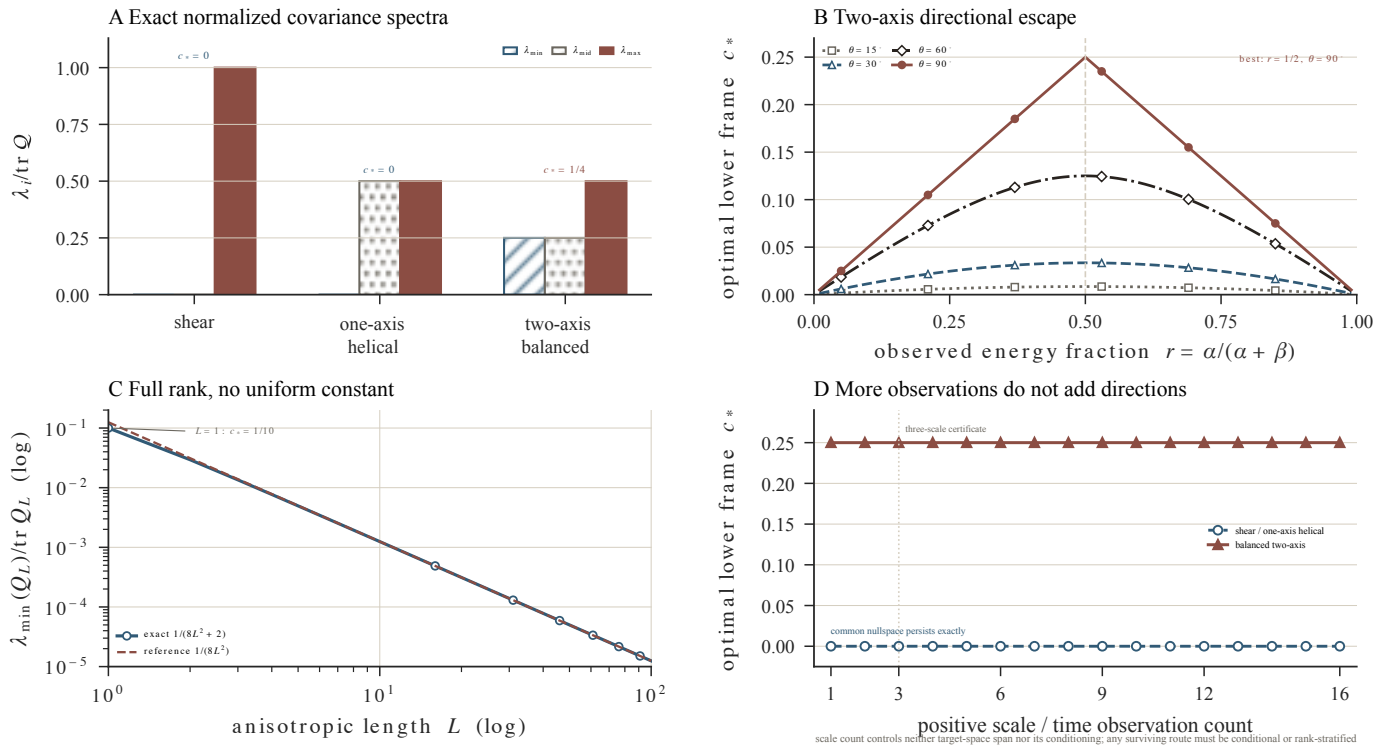


Figure Ro.70N. A: After normalization, the periodic shear has spectrum $(0, 0, 1)$, the single-axis helical wave has $(0, 1/2, 1/2)$, and the balanced two-axis positive control has $(1/4, 1/4, 1/2)$. B: The two-axis frame constant depends on both the angle between the axes and their energy balance. C: The whole-space Gaussian family remains full rank, but $c_* = 1/(8L^2 + 2)$ tends to zero. D: Repeatedly adding scale or time samples in the same direction does not increase rank. All data come from closed-form formulas, not DNS.

The value lies in rigorously ruling out a seemingly natural but invalid bridge

Multiscale energy coverage and a directional frame are two different things. A Littlewood–Paley or wavelet decomposition can fully record scalar energy, but it does not automatically mix directions in the vector target space. Ro.70N turns this distinction into a kernel formula, exact PDE counterexamples, and quantitative full-rank degenerating families.

This step does not reach a regularity theorem, but it has a clear routing value: analytic counterexamples have already ruled out universal coercivity, so there is no reason to use large DNS calculations to search for a uniform constant that does not exist. Computational resources should be reserved for propositions that have survived the algebraic gates and can connect to critical continuation criteria.

A bounded search of the primary literature through 2026-08-25 found no exactly matching no-go theorem for finite-scale vorticity Gramians; however, vorticity covariance, structure tensors, frames, persistent excitation, Beltrami decompositions, and anisotropic continuation criteria all have clear precedents. The supportable claim is therefore the audit result for this specific set of propositions, not the first introduction of these concepts.

Ro.70O turns to the coercive–near-plane–near-line trichotomy

Let $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq 0$ be the eigenvalues of the aggregated covariance. The next stage treats separately

$\lambda_3 \gtrsim \text{tr } Q,$	quantitative coercivity;
$\lambda_3 \ll \text{tr } Q,$	vorticity nearly planar;
$\lambda_2 + \lambda_3 \ll \text{tr } Q,$	vorticity nearly aligned with a line.

Only the near-line regime is geometrically close to the two-vorticity-component criteria of Chae–Choe and Miller. Two bridges are still missing: one from filtered, spatially averaged spectral degeneration to critical space-time norms of the unfiltered vorticity, and another providing a controllable ledger for the principal direction as it varies in space and time. Ro.70O will first state the exact propositions needed for these two bridges and then decide which branch merits computation.

What is closed is the universal scalar-frame route, not the low-rank regularity problem

PROVED Exact kernel and optimal constant The kernel of a nonnegative aggregate is the intersection of the kernels of the positively weighted single-scale terms.	PROVED Periodic NSE counterexamples The shear has rank one and the single-axis helical wave rank two; both are smooth unforced solutions.
---	--

PROVED

No uniform full-rank lower bound

Genuinely three-dimensional perturbations and the whole-space Schwartz family make c_* tend to zero.

OPEN

Rank stratification toward continuation

The bridge between the filtered spectrum and critical unfiltered norms has not yet been established.

This version does not claim that all Beltrami fields are rank-deficient; it does not rule out conditional frames, matrix-valued mixing filters, or augmented observations, nor does it claim that low covariance rank itself implies regularity.

No new continuation criterion is obtained at this stage. It proves neither blow-up nor global smoothness, and it does not solve the Millennium problem.

11 / REPRODUCIBILITY AND SOURCES

The exact generator, independent review, literature boundary, and formal figure package have all been archived

[View the complete mathematical report](#) · [View the bounded primary-literature audit](#) · [View the independent internal review](#)

[View the exact certificate and SHA-256 manifest](#) · [View the formal journal-figure package](#) · [下载同步 PDF \(Chinese PDF\)](#) · [View the Clay Mathematics Institute's official problem statement](#)

The producer certifies the finite-dimensional symbolic identities, eigenvalues, determinants, and Gaussian moments. The periodic-solution identities, common-subspace theorem, near-shear limit, and literature interpretation are separate analytic arguments; the independent review checked the nonlocal-filter hypotheses, pressure signs, whole-space boundary, and claim scope item by item.